

# Chapter 5: Second Order Ordinary Differential Equations

Lecture

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## Unit 5: Lecture Notes

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## Topic 28: Introduction — Homogeneous and Non-Homogeneous Equations

### 28.1 What is a Second Order ODE?

A **second order ordinary differential equation** is an equation involving an unknown function  $y(x)$ , its first derivative  $y'$ , and its second derivative  $y''$ .

**General Form:**

$$F(x, y, y', y'') = 0$$

or in **standard form** (solved for  $y''$ ):

$$y'' + P(x) y' + Q(x) y = R(x)$$

where  $P(x)$ ,  $Q(x)$ , and  $R(x)$  are functions of  $x$  (or constants).

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### 28.2 Classification

**Homogeneous Second Order ODE:**

$$y'' + P(x) y' + Q(x) y = 0$$

The right-hand side is **zero**. The equation has no external forcing term.

**Non-Homogeneous Second Order ODE:**

$$y'' + P(x) y' + Q(x) y = R(x), \quad R(x) \neq 0$$

The right-hand side  $R(x)$  is called the **forcing function**, **input**, or **non-homogeneous term**.

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### 28.3 The Superposition Principle

**Theorem:** If  $y_1(x)$  and  $y_2(x)$  are both solutions of the homogeneous equation:

$$y'' + P(x)y' + Q(x)y = 0$$

then  $c_1y_1 + c_2y_2$  is also a solution for any constants  $c_1, c_2$ .

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## 28.4 Linear Independence and the Wronskian

Two functions  $y_1$  and  $y_2$  are **linearly independent** if neither is a constant multiple of the other.

The **Wronskian** of  $y_1$  and  $y_2$  is:

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1y_2' - y_2y_1'$$

- If  $W \neq 0$  on an interval,  $y_1$  and  $y_2$  are linearly independent on that interval.
  - If  $W = 0$ , they are linearly dependent.
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## 28.5 Structure of the General Solution

For the **non-homogeneous** equation  $y'' + Py' + Qy = R(x)$ , the **general solution** is:

$$y = y_c + y_p$$

where: -  $y_c$  = **Complementary Function (CF)** = general solution of the associated homogeneous equation  $y'' + Py' + Qy = 0$  -  $y_p$  = **Particular Integral (PI)** = any one particular solution of the full non-homogeneous equation

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## 28.6 Identifying Homogeneous vs Non-Homogeneous

**Example set:**

| Equation                  | Type            | Reason                 |
|---------------------------|-----------------|------------------------|
| $y'' - 5y' + 6y = 0$      | Homogeneous     | RHS = 0                |
| $y'' + 4y' + 4y = e^x$    | Non-Homogeneous | RHS = $e^x \neq 0$     |
| $y'' - 3y' = 0$           | Homogeneous     | RHS = 0                |
| $y'' + y = \sin x$        | Non-Homogeneous | RHS = $\sin x \neq 0$  |
| $y'' + 2y' + y = x^2 + 1$ | Non-Homogeneous | RHS = $x^2 + 1 \neq 0$ |

## Topic 29: Complementary Functions (CF) — Homogeneous Solutions with Constant Coefficients

### 29.1 Setting Up the Characteristic Equation

For the homogeneous ODE with **constant coefficients**:

$$ay'' + by' + cy = 0 \quad (a \neq 0)$$

We try the solution  $y = e^{mx}$ . Then:

$$y' = me^{mx}, \quad y'' = m^2e^{mx}$$

Substituting:

$$am^2e^{mx} + bme^{mx} + ce^{mx} = 0$$

$$e^{mx}(am^2 + bm + c) = 0$$

Since  $e^{mx} \neq 0$ , we need:

$$am^2 + bm + c = 0$$

This is the **Characteristic Equation** (also called Auxiliary Equation).

### 29.2 Three Cases Based on Discriminant $\Delta = b^2 - 4ac$

The nature of roots  $m_1, m_2$  of the characteristic equation determines the form of CF:

| Case          | Roots                                  | Discriminant | CF Form                           |
|---------------|----------------------------------------|--------------|-----------------------------------|
| <b>Case 1</b> | Two distinct real roots $m_1 \neq m_2$ | $\Delta > 0$ | $y_c = c_1e^{m_1x} + c_2e^{m_2x}$ |

| Case          | Roots                                                     | Discriminant | CF Form                                                   |
|---------------|-----------------------------------------------------------|--------------|-----------------------------------------------------------|
| <b>Case 2</b> | Two equal<br>(repeated) real<br>roots<br>$m_1 = m_2 = m$  | $\Delta = 0$ | $y_c = (c_1 + c_2 x)e^{mx}$                               |
| <b>Case 3</b> | Two complex<br>conjugate roots<br>$m = \alpha \pm i\beta$ | $\Delta < 0$ | $y_c = e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$ |

### Case 1: Distinct Real Roots — Three Worked Examples

#### Example 29.1.1

**Problem:** Find the complementary function of  $y'' - 5y' + 6y = 0$ .

**Solution:**

**Step 1:** Write the characteristic equation.

$$m^2 - 5m + 6 = 0$$

**Step 2:** Factorise.

$$(m - 2)(m - 3) = 0$$

$$m_1 = 2, \quad m_2 = 3$$

**Step 3:** Since  $m_1 \neq m_2$  (distinct real roots), the CF is:

$$y_c = c_1 e^{2x} + c_2 e^{3x}$$

**Verification:** Differentiate  $y_c$  and substitute back:

$$y'_c = 2c_1 e^{2x} + 3c_2 e^{3x}$$

$$y''_c = 4c_1 e^{2x} + 9c_2 e^{3x}$$

$$\begin{aligned} y''_c - 5y'_c + 6y_c &= (4c_1 e^{2x} + 9c_2 e^{3x}) - 5(2c_1 e^{2x} + 3c_2 e^{3x}) + 6(c_1 e^{2x} + c_2 e^{3x}) \\ &= c_1 e^{2x}(4 - 10 + 6) + c_2 e^{3x}(9 - 15 + 6) = 0 + 0 = 0 \quad \square \end{aligned}$$


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### Example 29.1.2

**Problem:** Solve  $y'' - y' - 12y = 0$ .

**Solution:**

**Step 1:** Characteristic equation:

$$m^2 - m - 12 = 0$$

**Step 2:** Factorise:

$$(m - 4)(m + 3) = 0$$

$$m_1 = 4, \quad m_2 = -3$$

**Step 3:** Distinct real roots, so:

$$y = c_1 e^{4x} + c_2 e^{-3x}$$


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### Example 29.1.3

**Problem:** Solve  $2y'' + 3y' - 2y = 0$ .

**Solution:**

**Step 1:** Characteristic equation (divide by 2 or keep as is):

$$2m^2 + 3m - 2 = 0$$

**Step 2:** Use quadratic formula:

$$m = \frac{-3 \pm \sqrt{9 + 16}}{4} = \frac{-3 \pm \sqrt{25}}{4} = \frac{-3 \pm 5}{4}$$

$$m_1 = \frac{-3 + 5}{4} = \frac{2}{4} = \frac{1}{2}, \quad m_2 = \frac{-3 - 5}{4} = \frac{-8}{4} = -2$$

**Step 3:**

$$y = c_1 e^{x/2} + c_2 e^{-2x}$$


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## Case 2: Repeated Real Roots — Three Worked Examples

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### Example 29.2.1

**Problem:** Find the CF of  $y'' - 6y' + 9y = 0$ .

**Solution:**

**Step 1:** Characteristic equation:

$$m^2 - 6m + 9 = 0$$

**Step 2:** Factorise:

$$(m - 3)^2 = 0$$

$$m_1 = m_2 = 3 \quad (\text{repeated root})$$

**Step 3:** For repeated roots  $m = 3$ , the CF is:

$$y_c = (c_1 + c_2 x)e^{3x}$$

**Why this form?** If we used  $y = c_1 e^{3x} + c_2 e^{3x} = (c_1 + c_2)e^{3x}$ , this would only be one independent solution. The factor  $x$  is introduced to generate a second linearly independent solution.

**Verification:**

$$y = (c_1 + c_2 x)e^{3x}$$

$$y' = c_2 e^{3x} + 3(c_1 + c_2 x)e^{3x} = e^{3x}(3c_1 + c_2 + 3c_2 x)$$

$$y'' = e^{3x}[3(3c_1 + c_2 + 3c_2 x) + 3c_2] = e^{3x}(9c_1 + 6c_2 + 9c_2 x)$$

$$y'' - 6y' + 9y = e^{3x}[(9c_1 + 6c_2 + 9c_2 x) - 6(3c_1 + c_2 + 3c_2 x) + 9(c_1 + c_2 x)]$$

$$= e^{3x}[9c_1 + 6c_2 + 9c_2 x - 18c_1 - 6c_2 - 18c_2 x + 9c_1 + 9c_2 x]$$

$$= e^{3x}[(9 - 18 + 9)c_1 + (6 - 6)c_2 + (9 - 18 + 9)c_2 x] = 0 \quad \square$$

### Example 29.2.2

**Problem:** Solve  $y'' + 4y' + 4y = 0$ .

**Solution:**

**Step 1:** Characteristic equation:

$$m^2 + 4m + 4 = 0$$

**Step 2:** Factorise:

$$(m + 2)^2 = 0$$

$$m = -2 \quad (\text{repeated})$$

**Step 3:**

$$y = (c_1 + c_2 x)e^{-2x}$$

### Example 29.2.3

**Problem:** Solve  $4y'' - 4y' + y = 0$ .

**Solution:**

**Step 1:** Characteristic equation:

$$4m^2 - 4m + 1 = 0$$

**Step 2:** Factorise:

$$(2m - 1)^2 = 0$$

$$m = \frac{1}{2} \quad (\text{repeated})$$

**Step 3:**

$$y = (c_1 + c_2x)e^{x/2}$$

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### Case 3: Complex Conjugate Roots — Three Worked Examples

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#### Example 29.3.1

**Problem:** Solve  $y'' - 2y' + 5y = 0$ .

**Solution:**

**Step 1:** Characteristic equation:

$$m^2 - 2m + 5 = 0$$

**Step 2:** Use the quadratic formula:

$$m = \frac{2 \pm \sqrt{4 - 20}}{2} = \frac{2 \pm \sqrt{-16}}{2} = \frac{2 \pm 4i}{2} = 1 \pm 2i$$

So  $\alpha = 1, \beta = 2$ .

**Step 3:** Complex roots  $\rightarrow$  use exponential-trigonometric form:

$$y_c = e^x(c_1 \cos 2x + c_2 \sin 2x)$$

**Physical meaning:** This solution represents **damped oscillations** — the  $e^x$  factor describes growth (since  $\alpha = 1 > 0$ ) while the trigonometric part gives oscillation at frequency  $\beta = 2$ .

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### Example 29.3.2

**Problem:** Solve  $y'' + 6y' + 13y = 0$ .

**Solution:**

**Step 1:** Characteristic equation:

$$m^2 + 6m + 13 = 0$$

**Step 2:** Quadratic formula:

$$m = \frac{-6 \pm \sqrt{36 - 52}}{2} = \frac{-6 \pm \sqrt{-16}}{2} = \frac{-6 \pm 4i}{2} = -3 \pm 2i$$

So  $\alpha = -3, \beta = 2$ .

**Step 3:**

$$y_c = e^{-3x}(c_1 \cos 2x + c_2 \sin 2x)$$

**Physical meaning:** This represents **damped decaying oscillations** since  $\alpha = -3 < 0$ .

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### Example 29.3.3

**Problem:** Solve  $y'' + 9y = 0$  with initial conditions  $y(0) = 3, y'(0) = -6$ .

**Solution:**

**Step 1:** Characteristic equation:

$$m^2 + 9 = 0 \implies m^2 = -9 \implies m = \pm 3i$$

Here  $\alpha = 0, \beta = 3$ .

**Step 2:** General solution:

$$y = e^{0 \cdot x} (c_1 \cos 3x + c_2 \sin 3x) = c_1 \cos 3x + c_2 \sin 3x$$

**Step 3:** Apply initial conditions.

$$\text{At } x = 0: y(0) = c_1 \cos 0 + c_2 \sin 0 = c_1 = 3$$

So  $c_1 = 3$ .

$$y' = -3c_1 \sin 3x + 3c_2 \cos 3x$$

$$\text{At } x = 0: y'(0) = -3c_1(0) + 3c_2(1) = 3c_2 = -6$$

So  $c_2 = -2$ .

**Step 4:** Particular solution:

$$y = 3 \cos 3x - 2 \sin 3x$$

This represents **pure undamped oscillation** (no decay or growth since  $\alpha = 0$ ).

### 29.3 Summary of CF Determination

Write ODE  $\rightarrow$  Form Characteristic Equation  $am^2 + bm + c = 0$

|                              |                      |                                 |
|------------------------------|----------------------|---------------------------------|
|                              |                      |                                 |
| Compute $\Delta = b^2 - 4ac$ |                      |                                 |
| /                            |                      | \                               |
| $\Delta > 0$                 | $\Delta = 0$         | $\Delta < 0$                    |
| (Distinct)                   | (Repeated)           | (Complex)                       |
| $m \quad m \text{ real}$     | $m = m = m$          | $m = \pm i$                     |
|                              |                      |                                 |
| $c e^{(m x)} + c e^{(m x)}$  | $(c + c x) e^{(mx)}$ | $e^{(x)} [c \cos x + c \sin x]$ |

### Topic 30: Particular Integrals — Method of Undetermined Coefficients (Further Reading)

The **Method of Undetermined Coefficients** is used to find a particular integral (PI)  $y_p$  for the non-homogeneous equation:

$$ay'' + by' + cy = R(x)$$

**Core idea:** We guess the form of  $y_p$  based on the form of  $R(x)$ , then substitute back into the ODE to find the unknown coefficients.

**This method works when  $R(x)$  is one of:** - A polynomial:  $R(x) = x^n, x^2 + 3x, \dots$  etc. - An exponential:  $R(x) = e^{kx}$  - Sine or cosine:  $R(x) = \sin \beta x$  or  $\cos \beta x$  - Products of the above

### Topic 31: Particular Integrals — Variation of Parameters (Further Reading)

The Method of Variation of Parameters is a **general method** that works for any forcing function  $R(x)$  — including functions like  $\tan x$ ,  $\ln x$ ,  $\sec x$ , or  $\frac{1}{x}$ , which cannot be handled by undetermined coefficients.

**Given:**  $y'' + P(x)y' + Q(x)y = R(x)$

**Assumption:** The CF is  $y_c = c_1 y_1 + c_2 y_2$  where  $y_1, y_2$  are known.

**Key Idea:** Replace the constants  $c_1, c_2$  with functions  $u_1(x), u_2(x)$ :

$$y_p = u_1(x) y_1 + u_2(x) y_2$$

The functions  $u_1$  and  $u_2$  are found by solving:

$$u_1' = -\frac{y_2 \cdot R(x)}{W(y_1, y_2)}, \quad u_2' = \frac{y_1 \cdot R(x)}{W(y_1, y_2)}$$

where the **Wronskian** is:

$$W = W(y_1, y_2) = y_1 y_2' - y_2 y_1'$$

Then integrate:

$$u_1 = \int -\frac{y_2 R}{W} dx, \quad u_2 = \int \frac{y_1 R}{W} dx$$

And the PI is:

$$y_p = u_1 y_1 + u_2 y_2$$

**Note:** Write the equation in standard form (coefficient of  $y''$  must be 1) before applying this formula.

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## Topic 32: Complete Solutions and Initial Value Problems (IVPs)

### 32.1 The Complete Solution

Recall the structure:

$$y_{\text{general}} = y_c + y_p$$

where: -  $y_c = c_1 y_1 + c_2 y_2$  is the **complementary function** (CF) with two arbitrary constants  $c_1, c_2$  -  $y_p$  is the **particular integral** (PI) — a specific function with no arbitrary constants

The **complete solution** (also called the **general solution**) is obtained by combining CF and PI.

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### 32.2 Initial Value Problems (IVPs)

An **Initial Value Problem (IVP)** consists of a second-order ODE plus two conditions specified at the same point  $x = x_0$ :

$$y(x_0) = y_0, \quad y'(x_0) = y_1$$

**Procedure:** 1. Find the CF  $\rightarrow$  includes  $c_1, c_2$  2. Find the PI 3. Write complete solution:  $y = y_c + y_p$  4. Differentiate:  $y' = y'_c + y'_p$  5. Apply the two initial conditions to get two equations in  $c_1, c_2$  6. Solve for  $c_1$  and  $c_2$  7. Substitute back to get the unique particular solution

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**Example 32.1 — Exponential Forcing with IVP**

**Problem:** Solve the IVP:

$$y'' - 3y' + 2y = 4e^x, \quad y(0) = 1, \quad y'(0) = 0$$

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**Step 1: Find the CF**

Characteristic equation:  $m^2 - 3m + 2 = 0$

$$(m - 1)(m - 2) = 0 \implies m_1 = 1, \quad m_2 = 2$$

$$y_c = c_1 e^x + c_2 e^{2x}$$

---

**Step 2: Find the PI**

$R(x) = 4e^x$ . Note:  $a = 1$  is a root ( $m_1 = 1$ ), so modify.

**Trial:**  $y_p = Axe^x$

$$y'_p = Ae^x + Axe^x = Ae^x(1 + x)$$

$$y''_p = Ae^x + Ae^x(1 + x) = Ae^x(2 + x)$$

Substitute into ODE:

$$Ae^x(2 + x) - 3Ae^x(1 + x) + 2Axe^x = 4e^x$$

$$Ae^x[(2 + x) - (3 + 3x) + 2x] = 4e^x$$

$$Ae^x[2 + x - 3 - 3x + 2x] = 4e^x$$

$$Ae^x[-1 + 0 \cdot x] = 4e^x$$

$$-A = 4 \implies A = -4$$

$$y_p = -4xe^x$$

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**Step 3: Complete Solution**

$$y = c_1e^x + c_2e^{2x} - 4xe^x$$

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**Step 4: Derivative**

$$y' = c_1e^x + 2c_2e^{2x} - 4e^x - 4xe^x$$

$$y' = (c_1 - 4)e^x + 2c_2e^{2x} - 4xe^x$$

---

**Step 5: Apply initial conditions**

At  $x = 0$ :

$$y(0) = c_1e^0 + c_2e^0 - 4(0)e^0 = c_1 + c_2 = 1 \quad \dots (1)$$

$$y'(0) = (c_1 - 4)e^0 + 2c_2e^0 - 0 = c_1 - 4 + 2c_2 = 0 \quad \dots (2)$$

---

**Step 6: Solve for constants**

From (2):  $c_1 + 2c_2 = 4 \quad \dots (2')$

Subtract (1) from (2'):  $c_2 = 3$

Substitute in (1):  $c_1 + 3 = 1 \implies c_1 = -2$

---

### Step 7: Particular solution of IVP

$$y = -2e^x + 3e^{2x} - 4xe^x$$

**Verification:** At  $x = 0$ :  $y = -2 + 3 - 0 = 1 \quad \square$

$y' = -2e^x + 6e^{2x} - 4e^x - 4xe^x$ ; at  $x = 0$ :  $y' = -2 + 6 - 4 - 0 = 0 \quad \square$

---

### Example 32.2 — Polynomial Forcing with IVP

**Problem:** Solve the IVP:

$$y'' + 4y' + 4y = 8x^2 + 4x, \quad y(0) = -1, \quad y'(0) = 3$$


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#### Step 1: CF

Characteristic equation:  $m^2 + 4m + 4 = 0$

$(m + 2)^2 = 0 \implies m = -2$  (repeated)

$$y_c = (c_1 + c_2x)e^{-2x}$$


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#### Step 2: PI

$R(x) = 8x^2 + 4x$  (degree 2 polynomial). Trial:  $y_p = Ax^2 + Bx + C$

$$y'_p = 2Ax + B$$

$$y''_p = 2A$$

Substitute:

$$2A + 4(2Ax + B) + 4(Ax^2 + Bx + C) = 8x^2 + 4x$$

$$4Ax^2 + (8A + 4B)x + (2A + 4B + 4C) = 8x^2 + 4x + 0$$

**Comparing coefficients:**

$$x^2: 4A = 8 \implies A = 2$$

$$x^1: 8A + 4B = 4 \implies 16 + 4B = 4 \implies B = -3$$

$$x^0: 2A + 4B + 4C = 0 \implies 4 - 12 + 4C = 0 \implies C = 2$$

$$y_p = 2x^2 - 3x + 2$$


---

**Step 3: Complete Solution**

$$y = (c_1 + c_2x)e^{-2x} + 2x^2 - 3x + 2$$


---

**Step 4: Derivative**

$$y' = c_2e^{-2x} - 2(c_1 + c_2x)e^{-2x} + 4x - 3$$

$$y' = (-2c_1 + c_2 - 2c_2x)e^{-2x} + 4x - 3$$


---

**Step 5: Apply initial conditions**

At  $x = 0$ :

$$y(0) = c_1e^0 + 0 + 0 - 0 + 2 = c_1 + 2 = -1 \implies c_1 = -3 \quad \dots (1)$$

$$y'(0) = (-2c_1 + c_2)e^0 - 3 = -2c_1 + c_2 - 3 = 3 \quad \dots (2)$$

$$\text{From (2): } -2(-3) + c_2 = 6 \implies 6 + c_2 = 6 \implies c_2 = 0$$

---

**Step 6: Final Solution**

With  $c_1 = -3$ ,  $c_2 = 0$ :

$$y = -3e^{-2x} + 2x^2 - 3x + 2$$

**Verification:**

At  $x = 0$ :  $y = -3 + 0 - 0 + 2 = -1$   $\square$

$y' = 6e^{-2x} + 4x - 3$ ; at  $x = 0$ :  $y' = 6 - 3 = 3$   $\square$

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**Example 32.3 — Trigonometric Forcing with IVP**

**Problem:** Solve the IVP:

$$y'' + 9y = 18 \sin x, \quad y(0) = 2, \quad y'(0) = 3$$

---

**Step 1: CF**

Characteristic equation:  $m^2 + 9 = 0 \implies m = \pm 3i$

$$y_c = c_1 \cos 3x + c_2 \sin 3x$$

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**Step 2: PI**

$R(x) = 18 \sin x$ .  $\beta = 1$ , and  $\pm i$  are NOT roots of char. equation ( $m = \pm 3i$ ), so no modification.

Trial:  $y_p = A \cos x + B \sin x$

$$y'_p = -A \sin x + B \cos x$$

$$y''_p = -A \cos x - B \sin x$$

Substitute:

$$(-A \cos x - B \sin x) + 9(A \cos x + B \sin x) = 18 \sin x$$

$$8A \cos x + 8B \sin x = 18 \sin x$$

**Comparing:**

$$\cos x: 8A = 0 \implies A = 0$$

$$\sin x: 8B = 18 \implies B = \frac{9}{4}$$

$$y_p = \frac{9}{4} \sin x$$


---

**Step 3: Complete Solution**

$$y = c_1 \cos 3x + c_2 \sin 3x + \frac{9}{4} \sin x$$


---

**Step 4: Derivative**

$$y' = -3c_1 \sin 3x + 3c_2 \cos 3x + \frac{9}{4} \cos x$$


---

**Step 5: Apply initial conditions**

At  $x = 0$ :

$$y(0) = c_1(1) + c_2(0) + \frac{9}{4}(0) = c_1 = 2 \implies c_1 = 2 \quad \dots (1)$$

$$y'(0) = -3c_1(0) + 3c_2(1) + \frac{9}{4}(1) = 3c_2 + \frac{9}{4} = 3$$

$$3c_2 = 3 - \frac{9}{4} = \frac{12 - 9}{4} = \frac{3}{4} \implies c_2 = \frac{1}{4}$$


---

**Step 6: Final Solution**

$$y = 2 \cos 3x + \frac{1}{4} \sin 3x + \frac{9}{4} \sin x$$

**Verification:**

At  $x = 0$ :  $y = 2(1) + 0 + 0 = 2 \quad \square$

$y'(0) = 0 + \frac{3}{4} + \frac{9}{4} = \frac{12}{4} = 3 \quad \square$

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**Example 32.4 — Complete Solution using Variation of Parameters with IVP**

**Problem:** Solve the IVP:

$$y'' - y = 2x, \quad y(0) = 0, \quad y'(0) = 0$$


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**Step 1: CF**

$$m^2 - 1 = 0 \implies m = \pm 1$$

$$y_c = c_1 e^x + c_2 e^{-x}$$


---

**Step 2: PI by Undetermined Coefficients**

$R(x) = 2x$ . Trial:  $y_p = Ax + B$

$$y'_p = A, \quad y''_p = 0$$

Substitute:  $0 - (Ax + B) = 2x$

$$-A = 2 \implies A = -2; \quad -B = 0 \implies B = 0$$

$$y_p = -2x$$


---

**Step 3: Complete Solution**

$$y = c_1 e^x + c_2 e^{-x} - 2x$$


---

**Step 4: Derivative**

$$y' = c_1 e^x - c_2 e^{-x} - 2$$


---

**Step 5: Apply ICs**

At  $x = 0$ :

$$y(0) = c_1 + c_2 = 0 \quad \dots (1)$$

$$y'(0) = c_1 - c_2 - 2 = 0 \implies c_1 - c_2 = 2 \quad \dots (2)$$

Adding (1) and (2):  $2c_1 = 2 \implies c_1 = 1$

From (1):  $c_2 = -1$

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**Step 6: Final Solution**

$$y = e^x - e^{-x} - 2x$$

Note:  $e^x - e^{-x} = 2 \sinh x$ , so this can also be written as  $y = 2 \sinh x - 2x$ .

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**Example 32.5 — Complex Roots, Combined Forcing and IVP**

**Problem:** Solve the IVP:

$$y'' + 2y' + 5y = 4e^{-x} \cos 2x, \quad y(0) = 1, \quad y'(0) = -1$$


---

**Step 1: CF**

$$m^2 + 2m + 5 = 0$$

$$m = \frac{-2 \pm \sqrt{4 - 20}}{2} = \frac{-2 \pm 4i}{2} = -1 \pm 2i$$

$$y_c = e^{-x}(c_1 \cos 2x + c_2 \sin 2x)$$


---

**Step 2: PI**

$$R(x) = 4e^{-x} \cos 2x.$$

The “base” of the exponential is  $-1$  and the trig frequency is  $2$ , so we check whether  $-1 \pm 2i$  are roots. They ARE the characteristic roots!

**Modified trial:** Since  $e^{-x} \cos 2x$  and  $e^{-x} \sin 2x$  appear in CF, multiply by  $x$ :

$$y_p = xe^{-x}(A \cos 2x + B \sin 2x)$$

**Differentiate:** Let  $u = e^{-x}(A \cos 2x + B \sin 2x)$  so  $y_p = xu$ .

$$u = e^{-x}(A \cos 2x + B \sin 2x)$$

$$u' = e^{-x}[(-A + 2B) \cos 2x + (-2A - B) \sin 2x]$$

$$y_p' = u + xu'$$

$$y_p'' = 2u' + xu''$$

$$u'' = e^{-x}[(A - 4B) \cos 2x + (4A + B - 4A) \sin 2x]$$

Wait, let's compute  $u''$  carefully:

$$u = e^{-x}(A \cos 2x + B \sin 2x)$$

$$u' = -e^{-x}(A \cos 2x + B \sin 2x) + e^{-x}(-2A \sin 2x + 2B \cos 2x)$$

$$u' = e^{-x}[(-A + 2B) \cos 2x + (-B - 2A) \sin 2x]$$

$$u'' = e^{-x}[(A - 2B - 2(-A + 2B)) \cos 2x + (B + 2A - 2(-B - 2A)) \sin 2x]$$

$$= e^{-x}[(A - 2B + 2A - 4B) \cos 2x + (B + 2A + 2B + 4A) \sin 2x]$$

$$= e^{-x}[(3A - 6B) \cos 2x + (6A + 3B) \sin 2x]$$

$$= 3e^{-x}[(A - 2B) \cos 2x + (2A + B) \sin 2x]$$

Substitute  $y_p = xu$ ,  $y'_p = u + xu'$ ,  $y''_p = 2u' + xu''$  into the ODE:

$$y''_p + 2y'_p + 5y_p = (2u' + xu'') + 2(u + xu') + 5xu$$

$$= 2u' + 2u + x(u'' + 2u' + 5u)$$

But  $u$  satisfies the **homogeneous** equation (since  $e^{-x} \cos 2x$  is in CF):

$$u'' + 2u' + 5u = 0$$

Therefore:

$$y''_p + 2y'_p + 5y_p = 2u' + 2u$$

We need  $2u' + 2u = 4e^{-x} \cos 2x$ , i.e.,  $u' + u = 2e^{-x} \cos 2x$ .

$$u' + u = e^{-x}[(-A + 2B) \cos 2x + (-B - 2A) \sin 2x] + e^{-x}[A \cos 2x + B \sin 2x]$$

$$= e^{-x}[(2B) \cos 2x + (-2A) \sin 2x]$$

Set equal to  $2e^{-x} \cos 2x$ :

$$\cos 2x: 2B = 2 \implies B = 1$$

$$\sin 2x: -2A = 0 \implies A = 0$$

$$y_p = xe^{-x}(0 \cdot \cos 2x + 1 \cdot \sin 2x) = xe^{-x} \sin 2x$$


---

### Step 3: Complete Solution

$$y = e^{-x}(c_1 \cos 2x + c_2 \sin 2x) + xe^{-x} \sin 2x$$


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### Step 4: Derivative

$$y' = -e^{-x}(c_1 \cos 2x + c_2 \sin 2x) + e^{-x}(-2c_1 \sin 2x + 2c_2 \cos 2x) + e^{-x} \sin 2x + xe^{-x}(2 \cos 2x - \sin 2x)$$

At  $x = 0$ :

$$y'(0) = -(c_1) + (2c_2) + 0 + 0 = -c_1 + 2c_2$$


---

### Step 5: Apply ICs

At  $x = 0$ :

$$y(0) = c_1 = 1 \quad \dots (1)$$

$$y'(0) = -c_1 + 2c_2 = -1 \implies -1 + 2c_2 = -1 \implies c_2 = 0 \quad \dots (2)$$


---

### Step 6: Final Solution

$$y = e^{-x} \cos 2x + xe^{-x} \sin 2x$$

This can be written compactly as:

$$y = e^{-x}(\cos 2x + x \sin 2x)$$

**Physical Interpretation:** This represents a **resonant response** — the  $xe^{-x} \sin 2x$  term grows linearly with  $x$  before the exponential decay dominates. This is analogous to resonance in a damped mechanical or electrical system.

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## 32.3 Summary: Step-by-Step Procedure for Solving IVPs

STEP 1: Find CF

- Write characteristic equation
- Find roots  $m_1, m_2$
- Write  $y_c$  (3 cases)

STEP 2: Find PI

- Check form of  $R(x)$
- Choose method: Undetermined Coefficients or Variation of Parameters
- Check modification rule
- Solve for constants A, B, etc.
- Write  $y_p$

#### STEP 3: Complete Solution

- $y = y_c + y_p$

#### STEP 4: Differentiate

- Compute  $y'$

#### STEP 5: Apply Initial Conditions

- Substitute  $x = x$  in  $y$ : get equation for  $c, c$
- Substitute  $x = x$  in  $y'$ : get second equation
- Solve the system of 2 equations

#### STEP 6: Write Final Answer

- Substitute  $c, c$  back
- Always verify by substitution

### 32.4 Quick Summary Table for Complete Solutions

| Component                                    | Formula                                              | Role                                      |
|----------------------------------------------|------------------------------------------------------|-------------------------------------------|
| <b>CF: Distinct roots</b> $m_1 \neq m_2$     | $c_1 e^{m_1 x} + c_2 e^{m_2 x}$                      | Transient (general homogeneous behaviour) |
| <b>CF: Repeated root</b> $m_1 = m_2 = m$     | $(c_1 + c_2 x) e^{mx}$                               | Transient (critically damped)             |
| <b>CF: Complex roots</b> $\alpha \pm i\beta$ | $e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$ | Transient (oscillatory)                   |
| <b>PI: Undetermined coefficients</b>         | $y_p =$ trial form matching $R(x)$                   | Steady-state response                     |
| <b>PI: Variation of parameters</b>           | $y_p = u_1 y_1 + u_2 y_2$                            | For any $R(x)$                            |

| Component                      | Formula                               | Role            |
|--------------------------------|---------------------------------------|-----------------|
| <b>General solution</b>        | $y = y_c + y_p$                       | Full behaviour  |
| <b>Particular IVP solution</b> | $y$ with $c_1, c_2$ determined by ICs | Unique solution |

### Overall Summary of Unit 5 (Topics 28–32)

| Topic                                    | Core Concept                                         | Key Method/Formula                                              |
|------------------------------------------|------------------------------------------------------|-----------------------------------------------------------------|
| <b>28. Classification</b>                | Homogeneous vs Non-Homogeneous                       | $y'' + Py' + Qy = 0$ vs $R(x) \neq 0$                           |
| <b>29. CF</b>                            | Solve characteristic equation<br>$am^2 + bm + c = 0$ | 3 cases: distinct, repeated, complex roots                      |
| <b>30. PI: Undetermined Coefficients</b> | Guess form of $y_p$ based on $R(x)$                  | Modification rule when trial $\in$ CF                           |
| <b>31. PI: Variation of Parameters</b>   | $y_p = u_1y_1 + u_2y_2$                              | $u_i$ found using Wronskian — works for all $R(x)$              |
| <b>32. Complete Solution + IVP</b>       | $y = y_c + y_p$ , then apply ICs                     | 2 ICs $\rightarrow$ 2 equations $\rightarrow$ unique $c_1, c_2$ |